

SECTION A [20*0.5 = 10 Marks]

Write your answers in the table below

Q #	1	2	3	4	5	6	7	8	9	10
Answer	C	C	A	B	C	A	A	D	B	B
Q#	11	12	13	14	15	16	17	18	19	20
	D	C	D	B	C	31	$x^8 + x^4 + x^3 + x + 1$	D1	$GF(2^8)$	32

1. Which of the following statements about DES is FALSE?

- ☐ A. DES uses a Feistel structure
☒ C. S-Boxes are key-dependent
☐ B. DES operates on 64-bit blocks
☐ D. DES is not secure

2. If the same sub-key is used in all 16 rounds of DES

- ☐ A. Security improves
☒ C. Security significantly weakens
☐ B. Cipher becomes linear
☐ D. No effect

3. Which statement about weak keys in DES is TRUE?

- ☒ A. They produce identical round keys
☐ C. They bypass S-Boxes
☐ B. They reduce block size
☐ D. They skip rounds

4. Which of the following is required to construct $GF(2^n)$?

- ☐ A. Prime number
☒ B. Irreducible polynomial
☐ C. Even number
☐ D. Identity Matrix

5. Which of the following is NOT a DES property?

- ☒ A. Feistel Structure
☒ C. Variable block size
☐ B. Symmetric key
☐ D. Fixed S-Boxes

6. Triple DES (EDE mode) means:

- ☒ A. Encrypt-Decrypt-Encrypt
☐ C. Decrypt-Encrypt-Decrypt
☐ B. Encrypt-Encrypt-Decrypt
☐ D. Encrypt-Decrypt-Decrypt

SECTION B [10 Marks]

1. Consider the DES encryption algorithm with 16 rounds, where the round function $F(R, K)$ operates on a 32-bit input R and a 48-bit sub-key K .

Analyze the following two cases:

- Assume that the function $F(R, K)$ outputs a **32-bit string of zeros** for all inputs.
- Assume that the function $F(R, K)$ outputs the **same value as the input R** , ($F(R, K) = R$)

Answer following questions based on this information:

- For each case, derive the expressions for the final outputs L_{16} and R_{16} after 16 rounds of encryption, in terms of the initial inputs L_0 and R_0 . [2 Marks]

$$L_i = R_{i-1} \\ R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Space for Calculation:

(a)

$$1 \begin{cases} L_1 = R_0 \\ R_1 = L_0 \oplus \underbrace{F(R_0, K_1)}_0 \end{cases}$$

It converges after 2nd round
 $16 \bmod 2 = 0$

$$2 \begin{cases} L_2 = L_0 \\ R_2 = R_0 \end{cases} \quad L_{16} = L_0 \\ R_{16} = R_0$$

Answer:

(b) $F(R, K) = R$

$$1 \begin{cases} L_1 = R_0 \\ R_1 = L_0 \oplus R_0 \end{cases}$$

$$L_2 = L_0 \oplus R_0 \\ R_2 = R_0 \oplus L_0 \oplus R_0 = L_0 \quad (\because R_0 \oplus R_0 = 0)$$

$$3 \begin{cases} L_3 = L_0 \\ R_3 = L_0 \oplus R_0 \oplus R_0 = R_0 \end{cases}$$

It converges after 3rd round
 $16 \bmod 3 = 1$

After 1st round

$$L_{16} = R_0, R_{16} = L_0 \oplus R_0$$

- For each case, derive the expressions for the original inputs L_0 and R_0 in terms of the final outputs L_{16} and R_{16} during decryption. [2 Marks]

$$L_{i-1} = R_i \oplus F(L_i, K_i)$$

$$R_{i-1} = L_i$$

Space for Calculation:

Case (a) $L_{16} = L_0, R_{16} = R_0$

$$L_{15} = R_0 \oplus 0 = R_0$$

$$R_{15} = L_0$$

it will be same

$$\boxed{L_0 = L_{16} \\ R_0 = R_{16}}$$

Answer:

$$(R_0, L_0 \oplus R_0)$$

Case (b) $R_{15} = R_0$

$$L_{15} = L_0 \oplus R_0 \oplus R_0 = L_0$$

$$R_{14} = L_0$$

$$L_{14} = R_0 \oplus L_0$$

$$R_{13} = R_0 \oplus L_0$$

$$L_{13} = L_0 \oplus R_0 \oplus L_0$$

$$= R_0$$

After 3rd round it will converge.

$$R_0 = R_0 = L_{16}$$

$$L_0 = L_0 \oplus R_0$$

$$\boxed{R_0 = L_{16} \\ L_0 = R_{16} \oplus L_{16}} \quad \text{i.e.}$$

$$\underline{R_0 \oplus L_0 \oplus R_0}$$

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2. Write addition and multiplication table for GF (5).

[2 Marks]

Answer:

Addition

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Multiplication

*	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

3. Write all polynomials for GF (3^2).

[2 Marks]

Answer:

Total number of elements = 9

$$a_i \in GF(3) \Rightarrow a_i \in \{0, 1, 2\}$$

polynomials will be of the form

$$a_1x + a_0$$

All polynomials — $0, 1, 2, x, x+1, x+2, 2x, 2x+1, 2x+2$

4. In the first round of AES encryption, the state matrix obtained after the ShiftRows operation is given below:

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

Perform the MixColumns operation and compute the value of the element $S_{3,2}'$ (i.e., the element in the 3rd row and 2nd column of the resulting matrix). (Check appendix at page 6) [2 Marks]

Answer: A1

Space for Calculation:

$$S_{3,2}' = 03 S_{0,2} + S_{1,2} + S_{2,2} + 02 S_{3,2}$$

$$= (03) 4D \oplus 90 \oplus 4A \oplus (02) D8$$

$$(03) 4D = (02) 4D \oplus 4D$$

$$02 4D = \boxed{0} \overset{\text{No XOR}}{100} 1101$$

$$\text{left shift} = 1001 1010$$

$$(03) 4D = \begin{array}{r} 1001 1010 \\ 0100 1101 \\ \hline 1101 0111 \end{array}$$

$$\begin{array}{r} 1101 0111 \\ 1010 1100 \\ \hline 0111 1011 \\ 1001 0000 \\ \hline 1110 1011 \\ 0100 1010 \\ \hline 1010 0001 \\ \hline A1 \end{array}$$

$$02 D8$$

$$\begin{array}{r} D8 = \boxed{1} 101 1000 \\ \text{left shift} = 1011 0001 \\ \hline = 0001 1101 \\ 1010 1100 \\ \hline AC \end{array}$$